

Teaching AI the Way We Teach Data Science

Open, hands-on, built to scale

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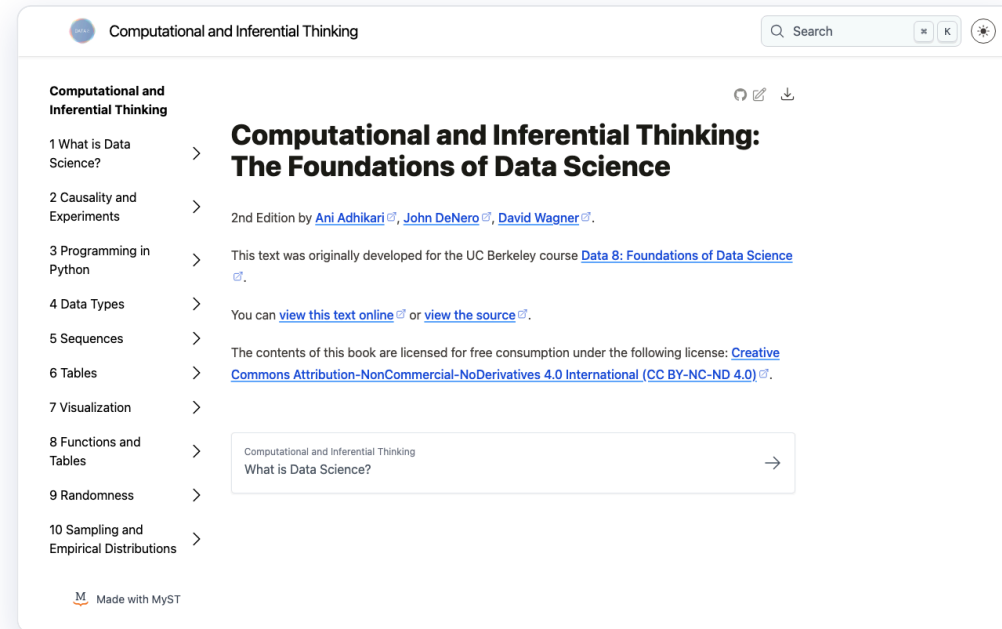
UC Berkeley College of Computing,
Data Science, and Society

The Data 8 Story



A first course in computing and statistics, for any major, in the browser, with real data from day one.

Largest course at UC Berkeley - up to 50% of undergraduates take it, and the gateway to Berkeley's biggest major.



inferentialthinking.com

Three lessons from a decade of scaling

1. WHAT DO WE KNOW WORKS?

Remove friction

JupyterHub means no installs and no laptop gap. Students meet ideas before they fight environments.

Connect to disciplines

Connector courses (like Data 88E) provide domain context that makes methods stick.

Open everything

Github Deployment, Open Stack of tools, Open textbook, notebooks, and autograding are what let it travel.

Scale is more than course materials

1. WHAT DO WE KNOW WORKS? HOW TO GET 25 COMMUNITY COLLEGES TO ADOPT IT?

- **Infrastructure:** shared JupyterHubs through for less resourced institutions (eg NSF Cloudbank or Cal-ICOR)
- **Curriculum:** open materials community colleges can adopt and adapt
- **People:** faculty networks, workshops training, and credit that transfers

How much of this can we do for AI education?

AI Education – how to keep the playing field level

2. CAN WE ADAPT WHAT WE TEACH?

- Some iterations of AI literacy risk shrinking into prompting tips and tricks
- Teaching through a rented chatbot teaches a **product**, not a **field**
- Closed access re-creates equity gaps: some students/institutions can pay, others cannot

Small Open Models as teaching tools

2. CAN WE ADAPT WHAT WE TEACH?

A small open model is just a file students can count, inspect, and test.

- Parameters $\times$ bits $\div$ 8 = size
- Tokens and next-token probabilities
- Temperature: what "randomness" means
- The linear algebra underneath
- "harness" - feed data in, get data out with data manipulation

```
from transformers import AutoTokenizer, AutoModelForCausalLM

name = "HuggingFaceTB/SmolLM2-135M" # a 270 MB file
tok = AutoTokenizer.from_pretrained(name)
lm = AutoModelForCausalLM.from_pretrained(name)

ids = tok("Amsterdam is famous for its", return_tensors="pt")
probs = lm(**ids).logits[0, -1].softmax(-1)

for p, i in zip(*probs.topk(5)):
    print(tok.decode(i), f"{p:.0%}")
```



Small Models in a Jupyter Hub

2. CAN WE ADAPT WHAT WE TEACH?

```
In [10]: # set the model path
model_path = "shared-rw/qwen2-1.5b-instruct-q4_0.gguf"
# example model path, change to your model location
```

```
In [11]: # Load the model from shared storage
model = Llama(
    model_path=model_path,
    n_ctx=2048, # Context window size (how much text the model can process at once)
    verbose=False # Set to True if you want to see detailed loading information
)

print("Model loaded successfully!")
```

Model loaded successfully!

Generating Text

Now let's try generating some text. We'll give the model a prompt and see what it produces.

```
In [12]: # Generate text from a simple prompt
import textwrap

prompt = "Python is a programming language that"

output = model(
    prompt,
    max_tokens=100, # Maximum length of generated text
    temperature=0.7, # Controls randomness (0 = deterministic, 1+ = creative)
    stop=["\n\n"] # Stop generation at double newline
)

# Extract the generated text
generated_text = output['choices'][0]['text']
print(f"Prompt: {prompt}")
print(f"Generated:\n{textwrap.fill(generated_text, width=80)}")
```

Prompt: Python is a programming language that
Generated: allows you to do many different things. It can be used for web development, data analysis, machine learning, and more. It's also a great language for beginners because it's easy to learn and has a lot of great resources online to help you get started. Python is also widely used in scientific research, as it's a powerful tool for data analysis and machine learning.

The model downloads into a shared location on *JupyterHub* and runs in browser. Students can inspect the model, change parameters, and see what happens, e.g. with no setup.

Admin sets up environment, prof sets up curriculum, student explores.

[Cloudbank Classroom](https://cloudbank.org/training/access-cloudbank-classroom): cloudbank.org/training/access-cloudbank-classroom

Small Models in the Browser

2. CAN WE ADAPT WHAT WE TEACH?



Try it now
jupyterlite.github.io/ai



The whole notebook, and the model, runs on your own laptop. **No account. No server. No cost.**

- **jupyterlite-ai**: chat and code completion inside JupyterLite
- **Pyodide**: Python compiled to WebAssembly
- **WebLLM**: LLMs on your GPU through WebGPU
- **Transformers.js**: Hugging Face models in the browser
- **willama**: llama.cpp compiled to WebAssembly

Thanks to Jeremy Tuloup!

Same notebook, three rungs

3. WHAT INVESTMENTS DO WE NEED?

1

Browser

JUPYTERLITE

Small models run on the student's own laptop. Free, private, zero setup.

2

Campus hub

JUPYTERHUB

Shared model folders, curated environments, GPUs when a course needs them.

3

HPC and cloud slices

JUPYTER AS THE FRONT DOOR

A small allocation for every student who needs one. Powerful but hard to use today, so **invest in making it easy**.

Teach the stack along the way: environments, containers, git, job schedulers, cloud costs. DevOps is part of AI literacy.

The model question

3. WHAT INVESTMENTS DO WE NEED?

- Today's small open models come mostly from a few labs, mostly China
- Corporate openness can change with the next release (e.g. OpenAI, Google, Meta)
- Build open curriculum that is flexible enough to swap models in and out, but worth having a small open model that is always available for teaching

Europe has pieces at every rung

3. WHAT INVESTMENTS DO WE NEED?

Rung	European pieces
Browser	QuantStack, JupyterLite
Campus hub	SURF and national research infrastructure
Public compute	EuroHPC
Models	GPT-NL, OpenEuroLLM

Plus a mandate: AI literacy under Article 4 of the EU AI Act

Two Goals

3. WHAT INVESTMENTS DO WE NEED?

A small, fully open,
multilingual **teaching
model** with documented
data

Shared open **curriculum**
that runs at multiple levels
and can move between
universities

Data Science to AI

We are still scaling data science.

Next: give everyone the ability to look inside AI.

For the panel: who should own the infrastructure that teaches the next generation what AI is?

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